AI-Powered Personalized Dietary Recommender for Insulin Resistance Management

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***Abstract*—Insulin resistance (IR) is a major risk factor for type 2 diabetes and cardiovascular disease, yet day-to-day dietary management remains largely manual and generic. While recent advances have produced AI-based tools for glucose prediction and metabolic modeling, current systems are typically static, population-level, and do not close the loop between real-time glucose feedback and personalized dietary recommendations. Moreover, explainability and privacy are often treated as afterthoughts, limiting user trust and large-scale deployment. This paper proposes an AI-powered, privacy-aware dietary recommendation system for adults with insulin resistance that integrates continuous glucose monitoring (CGM), meal logs, and lifestyle data. The system combines supervised models for postprandial glycemic response prediction with a reinforcement learning (RL) policy for adaptive meal planning, multi-modal data fusion for richer context, and explainable AI techniques to improve user understanding and adherence. Privacy is enforced through federated learning and differential privacy. We outline the data pipeline, model architecture, learning and deployment strategy, and an evaluation plan based on both machine learning metrics (MAE, RMSE) and clinical endpoints (time-in-range, HOMA-IR). The proposed framework aims to bridge scientific gaps in real-time, personalized, and privacy-preserving nutritional decision support for insulin resistance management.**

***Index Terms*—Insulin resistance, personalized nutrition, reinforcement learning, continuous glucose monitoring, explainable AI, federated learning, differential privacy.**

I. Introduction

Insulin resistance (IR) is a metabolic condition in which cells become less responsive to insulin, leading to elevated blood glucose levels and increased risk of type 2 diabetes and cardiovascular disease. Lifestyle interventions, particularly dietary modification, are a cornerstone of IR management. However, most individuals still rely on generic dietary advice and manual self-monitoring, which are not tailored to their unique metabolic responses.

In parallel, AI systems can now decode genomes, predict clinical risk, and interpret high-dimensional sensor data. Yet the application of AI to everyday nutrition is fragmented. Existing AI-driven nutrition applications often focus on simple calorie tracking, coarse meal scoring, or population-level guidelines. They rarely integrate continuous glucose monitoring (CGM), detailed meal logs, and lifestyle signals (e.g., activity, sleep) into a unified adaptive framework. Even when predictive models are used, the recommendations are typically static and not updated based on real-time glucose feedback. Furthermore, most systems are black boxes and do not explain why a recommendation is made, which can reduce user trust and adherence over time. Finally, the intensive collection of granular health and dietary data raises significant privacy concerns.

A. Problem Statement and Impact

Practical impact: The absence of a personalized and adaptive dietary recommender limits individuals' ability to manage daily glucose fluctuations and improve insulin sensitivity in a systematic way.

Scientific gap: To the best of our knowledge, no prior work has jointly unified:

- multimodal data fusion of CGM, food logs, and activity/sleep data,
- reinforcement learning for adaptive meal recommendation,
- explainable AI for transparent decision support, and
- privacy-preserving training through federated learning and differential privacy

into a single, deployable framework for insulin resistance management. [1]

B. Research Questions

We formalize our investigation through the following research questions (RQs):

- **RQ1:** Which machine learning or deep learning model most accurately predicts postprandial glycemic response based on meal composition and contextual factors?
- **RQ2:** Which reinforcement learning framework most effectively enables adaptive, personalized dietary recommendations based on real-time glucose feedback?
- **RQ3:** How can multimodal data—including CGM, food logs, and activity tracking data—be fused into a unified model to enhance real-time glucose prediction and dietary recommendation quality?
- **RQ4:** Which explainability techniques most effectively improve user trust and compliance in AI-based nutrition systems?

[1]Concepts and structure are based on our prior assignment writeup on AI-powered dietary recommendation for insulin resistance.

- **RQ5:** How can AI-based nutrition systems deliver personalized recommendations while preserving user privacy?
- **RQ6:** How can federated learning and differential privacy be leveraged in AI-based monitoring systems for IR to balance accuracy and privacy?

C. Contributions

This paper makes the following contributions:

1) We propose a multimodal AI framework that combines CGM traces, meal logs with nutrient breakdown, and wearable-derived activity/sleep metrics to predict post-prandial glucose response.
2) We design an RL-based recommendation layer that uses glucose outcomes as feedback to adapt meal suggestions over time, initially trained in simulation and then refined in real-world use.
3) We integrate explainable AI (XAI) methods, including feature attributions and counterfactual suggestions, to increase transparency and user trust.
4) We incorporate federated learning with client-side differential privacy and secure aggregation to protect user data while maintaining model performance.
5) We define an evaluation protocol that links technical metrics (MAE, RMSE) with clinically relevant endpoints (time-in-range, HOMA-IR) and user-centered measures (trust, adherence).

II. Related Work

A. Insulin Resistance Prediction and Diabetes Risk

Machine learning techniques have been applied to predict insulin resistance and diabetes risk using demographic, biochemical, and lifestyle factors [1], [4]. Gradient boosting and random forests have shown strong performance on structured clinical datasets for classifying IR status and predicting progression to type 2 diabetes.

B. AI for Personalized Nutrition

Recent work in personalized nutrition uses CGM data and food logs to model individual glycemic responses to meals [2], [**?**]. Neural networks and deep learning models have been used to predict postprandial spikes, while some systems attempt to optimize dietary patterns at a population level. However, most solutions remain static; they do not employ closed-loop adaptation using real-time feedback.

C. Digital Twins and Metabolic Simulation

Digital Twin frameworks simulate a virtual replica of an individual's metabolism and can test nutritional interventions in silico before real deployment [3]. These approaches provide a promising environment for testing and pre-training control policies but are not yet seamlessly integrated with real-world meal recommenders.

D. Reinforcement Learning for Glucose Control

RL has been successfully explored in automated insulin delivery and glucose control, particularly in type 1 diabetes [6], [7]. These works demonstrate that RL can learn policies to keep glucose within a target range under uncertain dynamics. However, most focus on insulin dosing rather than dietary decision-making, and target type 1 diabetes rather than insulin resistance and prediabetes.

E. Explainable AI in Nutrition and Health

Explainable AI techniques such as SHAP and LIME have been used to interpret diet and health prediction models [8]. Feature-level explanations and counterfactual suggestions can help users understand which meal components drive glucose changes and may improve adherence.

F. Privacy-Preserving Health AI

Federated learning and differential privacy are increasingly used to train models on sensitive data without centralizing raw records [9], [10]. In nutrition and wearable sensing, these techniques are still emerging but are well aligned with user expectations for confidentiality.

III. System Overview

A. Target Population

We focus on adults diagnosed with insulin resistance, identified by a HOMA-IR score of at least 2.5. Participants are recruited from outpatient clinics and screened for eligibility using standard clinical criteria.

B. Data Sources

Our system uses three primary data sources:

- **Continuous Glucose Monitoring (CGM):** Glucose values at 5-minute resolution from commercial CGM sensors (e.g., interstitial glucose) and from public resources such as the PhysioNet CGM macros dataset.
- **Meal Logs:** Timestamped records of meals captured via a smartphone diet-tracking app, with nutrient breakdown (carbohydrates, protein, fat, fiber, and energy).
- **Activity and Sleep:** Wearable-derived metrics including steps, heart rate, activity intensity, and sleep duration, summarized around meal events.

C. Study Design and Dataset

We aim for $N = 100$ participants (pilot $N = 30$), monitored over 12 weeks. For each participant, we collect:

- 5-minute CGM traces throughout the study period,
- detailed meal logs,
- activity and sleep metrics, and
- baseline and post-intervention HOMA-IR measurements.

Public CGM data (e.g., PhysioNet) are used to pre-train models and to construct simulated environments for RL training.

IV. Data Processing Pipeline

A. Temporal Alignment and Cleaning

All signals are aligned in time:

- CGM data are resampled to 5-minute bins.
- Missing CGM values over gaps ≤ 30 minutes are linearly interpolated.
- Days with more than 20% missing CGM data are excluded.

B. Nutrient Encoding

Each meal is converted into a numeric nutrient vector, normalized per 100 kcal: grams of carbohydrates, protein, fat, and fiber, plus optional micronutrient features where available. Additional categorical features (e.g., meal type: breakfast, lunch, dinner) are one-hot encoded.

C. Context Features from Wearables

For each meal, we extract wearable metrics in a window of ± 2 hours around the meal time:

- average and maximum heart rate,
- total steps and intensity levels, and
- sleep duration in the preceding night.

All numeric features are standardized (z-score) per participant to focus on intra-individual patterns.

D. Data Augmentation

To improve robustness and reduce overfitting, we apply simple augmentation strategies to the training data:

- jitter meal timestamps by $\pm$10–30 minutes,
- perturb portion sizes by $\pm 10\%$, and
- randomly drop low-impact snacks while preserving energy balance constraints.

V. Predictive Modeling

A. Prediction Task

Given a meal and the preceding context, the prediction model estimates the postprandial glucose trajectory over 0–240 minutes and derived scalar endpoints such as peak glucose and incremental area under the curve (iAUC).

B. Model Candidates

We compare three model families:

1) **Gradient-Boosted Trees (XGBoost):** Using engineered features (e.g., rolling glucose statistics, meal nutrients, activity summaries) as inputs.
2) **Sequence Model (LSTM):** An LSTM over recent CGM history with concatenated meal embeddings.
3) **Multimodal Temporal Model:** A 1D-CNN encoder for CGM sequences, a feedforward encoder for meal and activity features, and a small Transformer block for temporal fusion.

C. Training and Evaluation

We use participant-level cross-validation to avoid data leakage. Baselines include:

- a persistence model (predict future glucose as the last observed value),
- a simple feedforward neural network based on prior work [5].

Performance is assessed using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) on predicted glucose values, as well as error on peak glucose and iAUC.

VI. Reinforcement Learning for Meal Recommendation

A. Formulation

We formulate meal recommendation as a Markov Decision Process (MDP):

- **State:** Recent CGM history, upcoming time-of-day, current activity context, and a set of candidate meals consistent with user preferences.
- **Action:** Selection of one meal (or meal modification) from the candidate set.
- **Reward:** A function penalizing large glucose excursions (e.g., high 2-hour postprandial values, iAUC above threshold) while respecting user preferences and nutritional adequacy.

B. RL Algorithms

We plan to explore:

- Deep Q-Learning (DQN) with a discretized meal action space,
- Proximal Policy Optimization (PPO) for continuous modification of meal attributes (e.g., carbohydrate reduction),

using the predictive model as an environment simulator on historical CGM data before deployment.

C. Safety and Simulation

To mitigate risk, RL training is initially conducted offline in a simulated environment, using historical data and the learned glucose predictor as a surrogate environment. Only after policy stability and safety constraints are verified will limited-scope online adaptation be allowed.

VII. Privacy-Preserving Deployment

A. Federated Learning

Model updates are trained on-device using local user data, and only parameter updates or gradients are sent to a central aggregator. We adopt a FedAvg-like paradigm, where:

- clients perform several local optimization steps on their own data,
- the server aggregates updates into a global model,
- periodic personalization layers (e.g., final dense layers) are kept local.

B. Differential Privacy and Secure Aggregation

To further protect user confidentiality:

- Gaussian noise is added to gradient updates on the client side, enforcing user-level differential privacy;
- secure aggregation protocols ensure that the server can only observe aggregated updates, not individual contributions.

We perform an ablation study to compare centralized training, non-private federated learning, and DP-federated learning in terms of accuracy and privacy guarantees.

VIII. Explainability and User Interaction

A. Feature Attributions

We use SHAP-based feature attributions on the glucose prediction and recommendation models to highlight how each component of a meal (carbohydrates, protein, fat, fiber) and contextual features (e.g., prior glucose level, activity) contribute to predicted glucose outcomes.

B. Counterfactual Explanations

Beyond feature scores, the system generates actionable counterfactuals, such as:

> "If you reduce carbohydrates by 20 g and increase fiber by 5 g, your predicted 2-hour glucose peak decreases by X mg/dL."

C. User Study

We plan a user study measuring:

- perceived trust and understanding using 5-point Likert scales,
- acceptance rate of recommended meals,
- changes in dietary patterns over the intervention period.

IX. Evaluation Plan

A. Technical Metrics

We evaluate the predictive component with:

- MAE and RMSE on glucose trajectories,
- error on peak glucose and iAUC,
- calibration plots comparing predicted vs. observed outcomes.

B. Clinical Metrics

Clinical endpoints include:

- time-in-range (70–180 mg/dL) over the study period,
- improvement in HOMA-IR between baseline and follow-up,
- reduction in frequency of high postprandial spikes.

C. Ablation Studies

We perform ablations on:

- multimodal fusion (with vs. without wearable data),
- RL adaptation (static vs. adaptive policy),
- privacy layers (centralized vs. federated vs. DP-federated),
- explainability (with vs. without XAI in user interface) and its impact on adherence.

X. Ethical Considerations

All participants will provide informed consent. Data collection and processing will follow applicable ethical guidelines and regulations for digital health research. Users will have control over data sharing settings and may opt out of federated updates. Clear disclaimers will state that the system is a decision-support tool and not a replacement for professional medical advice.

XI. Conclusion and Future Work

We presented a design for an AI-powered, personalized dietary recommendation system targeting adults with insulin resistance. The proposed framework unifies multimodal data fusion, supervised prediction of postprandial glycemic response, reinforcement learning for adaptive meal recommendation, explainable AI for transparency, and privacy-preserving federated learning.

Future work includes implementing a full-scale Digital Twin component to simulate longer-term metabolic adaptation, extending the system to other metabolic conditions (e.g., type 2 diabetes and gestational diabetes), and integrating richer behavioral signals such as stress and mood. With sufficient validation and regulatory alignment, such systems could become central tools for scalable, personalized metabolic health management.

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